ENEX12002

LECTURE NOTES

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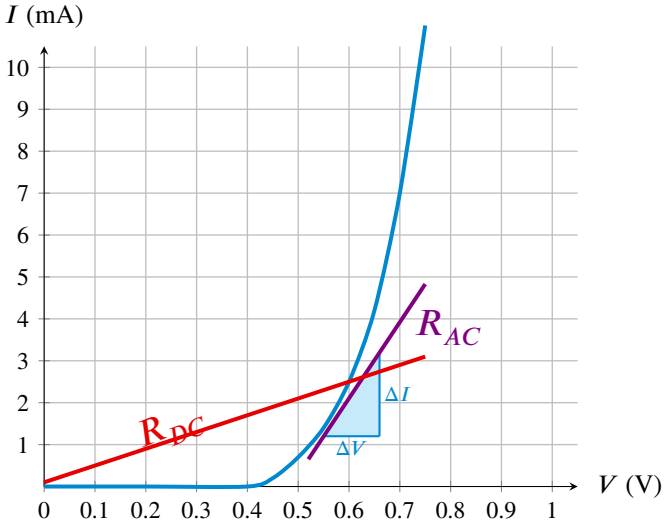
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1 Analog Circuits

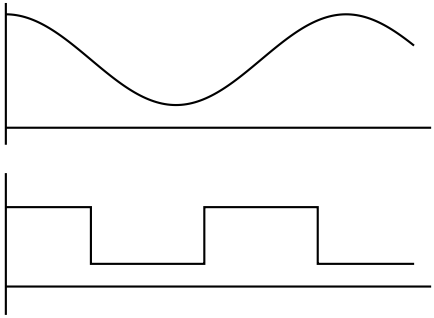
1.1 Characteristic Curves:

A characteristic curve is a type of linear plot that demonstates a relationship between two variable properties of a device. Most electronic devices CC refers too a plot of current as a function of coltage. Mostly known as an I-V curve.



1.2 Analog vs Digital Signals:

Signalling within a circuit is generally classified as either continuous or discrete. Continuous (Analog signalling) varies after a range of values within limits. Discrete (Digital signals) change between two discreet voltage levels.

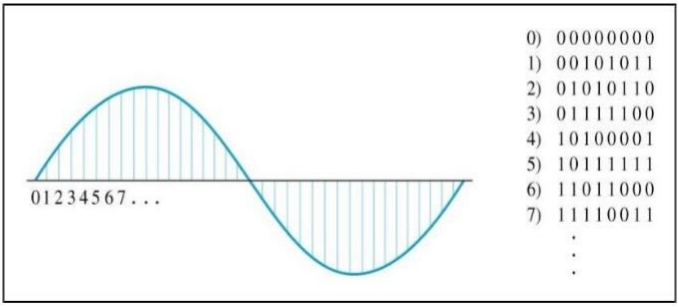


1.3 Analog & Digital Circuits:

Analog circuits are used in situations implementing linear functions, wave shaping and transformation between current and voltage. Digital circuit however ase used to process data rapidly and in calculations. Most systems utilize a mix of both systems (e.g AV systems).

1.4 Sampling:

When converting between signal types sampling must take place. Sampling is the approximation of analog's original waveform into time slices. The disadvantage of sampling is that is loses some precision, however the advantages of a digital system out weight this. The physical process of sampling is called quantizing. This produces numbers that can be processed by digital systems.



1.5 Vectors:

A vector is a line that contains both direction and magnitude. An AC circuits sinusodial waveform can be generated by plotting the project of a rotating vector in circular motion.

1.6 Frequency and Period:

The frequency of a periodic wave is the number of cycles that occur every second. The SI unit for this is Hz. The period is how long it takes a waveform to complete one cycle.

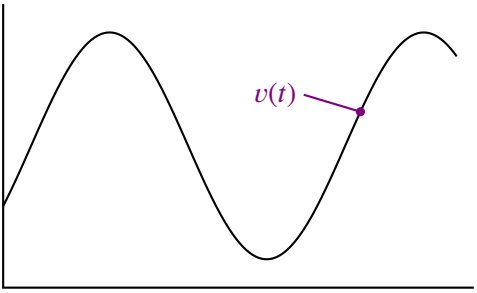
$$f = \frac{1}{T}$$

$$T = \frac{1}{f}$$

1.7 Instantaneous Values:

$$v(t) = V_p \sin(\omega t \pm \phi)$$

The instantaneous values are always against time (t).



1.8 Waveform Values:

The amplitude is the maximum distance of the values from the line of equilibrium. The average value of a sine wave is the found in $|v(t)|$, then averaging the curve.

$$V_{ave} = \frac{2V_p}{\pi} \approx 0.637V_p$$

1.9 RMS Values:

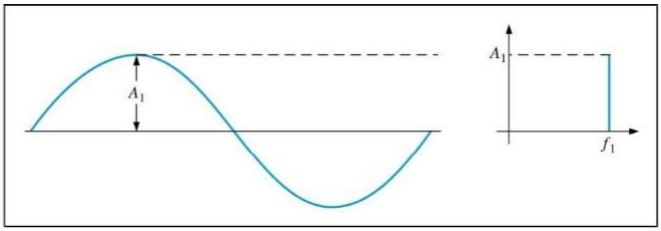
In order to compare AC and DC currents and voltages, AC is converted into its DC equivalent DC heating value. RMS current and voltage are their peak values.

$$\begin{aligned} V_{RMS} &= \frac{V_p}{\sqrt{2}} \\ &\approx 0.707V_p \end{aligned}$$

$$\begin{aligned} I_{RMS} &= \frac{I_p}{\sqrt{2}} \\ &\approx 0.707I_p \end{aligned}$$

1.10 Domain Signals:

Time domain is the funcation against time. An example is a osccilisope which shows values over a period of time. The frequency domain - A sine-wave against time can be represented against its frequency. This representations shows the waveforms amplitude and frequency but not it's phase angle.



1.11 Transducers:

These are a device that converts a physical quantity (e.g pressure, position or temp) from one form to another. Their outupts are quite small and require amplification. Passice transducers require external power to functions, this is called excitation. Active transducers are self-generating that convert a portion being measured into electrical signal.

1.12 Amplification:

Amps are designde to amplify either voltage or power. Linear amplifiers produce a magnified replica of the inputted signal, producing a useful output.

$$A_v = \frac{V_{out}}{V_{in}}$$

1.13 Transfer Curves:

All amplifiers, when ideal, produce a linear transfer curves. Beyond this limit they no longer act idealy, this is called saturation.

1.14 Power Gain:

$$A_p = \frac{P_{out}}{P_{in}} = \frac{V_{out}I_{out}}{V_{in}I_{in}}$$

1.15 dB Power Ratios:

The decibal of a power ratio is:

$$\begin{aligned} A_p &= 10\log_{10}(\frac{P_1}{P_2}) \\ &= 10\log_{10}(A_p) \end{aligned}$$

1.16 dB Voltage Ratio:

$$\begin{aligned} A_v &= 20\log_{10}(\frac{V_1}{V_2}) \\ &= 20\log_{10}(A_v) \end{aligned}$$

2 Basic Circuit Theory

2.1 Electrical Charge:

Electrical effects are attributed to the seperation of charges and their movements. Electrical forces are that opposites attract and similar repels. A charge is considered continuous if $q = 1.6022E\text{-}19e^{-}$.

2.2 Voltage and Current:

Rather than working purely in charge, it is easier to work with it in voltage and current.

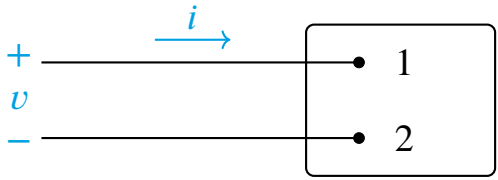
$$V = \frac{dw}{dq}$$

$$I = \frac{dq}{dt}$$

2.3 Ideal Basic Circuits:

These are made up of 5 main points:

1. Two Terminals
2. Can be described mathmatically in terms of voltage and current
3. Cannot be subdivided into further elements
4. Utilizes the passive sign convention

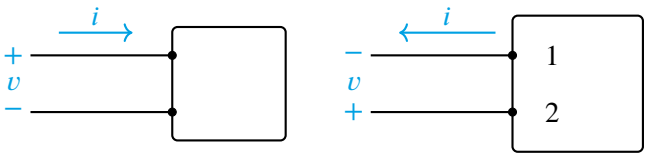


2.4 Power and Energy:

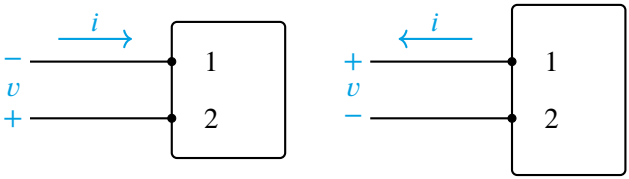
The useful output of a system is often non-electrical. Practical applications also have limitations in the amount of power it can handle.

$$p = \frac{dw}{dt} = (\frac{dw}{dq})(\frac{dq}{dt}) = VI$$

The polarity describes where current flows in the relationship to voltage. If $p > 0$, power is being delivered to the circuit inside the box. For instance a motor.



Else if $p < 0$, power is being extracted from a circuit. For example a battery.



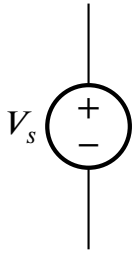
Elements that make up a basic circui include:

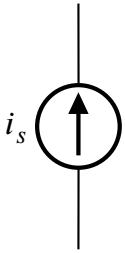
- Voltage source/s
- Current source/s
- Passive Circuit elements, As they do not produce electricity
  - Resistors
  - Inductors
  - Capacitors

2.5 Voltage and Current Sources:

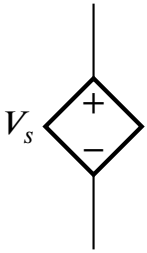
An electrical source that can convert between electrical and non-electrical energy. An ideal voltage source provides a consistant voltage regardless of current. This is similar for current sources. In real-world applications, ideal power sources do not exist.

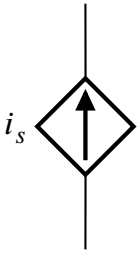
2.5.1 Source Types





Independent Sources





Dependent Sources

2.6 Kirchoffs Laws:

A circuit is said to be solved once a voltage and current is found across every circuit element.

$$\text{KCL} \Rightarrow \sum I_{out} = \sum I_{in}$$

$$\text{KVL} \Rightarrow \sum V_{loop} = 0$$

To apply these laws, polarity must be applied throughout the circuit.

2.7 Resistance:

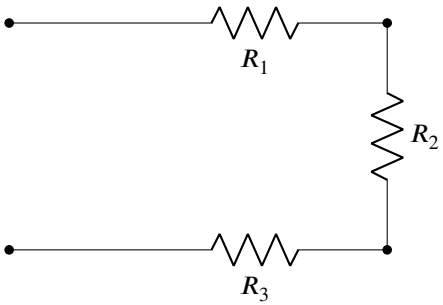
Resistors resist the flow of current within a circuit. Conductance is also the inverse of resistance, $G = \frac{1}{R}$.



$$V = IR$$

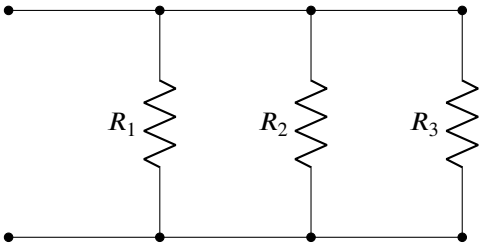
2.7.1 Series

$$R_{eq} = \sum_{n=1}^{k} R_n$$



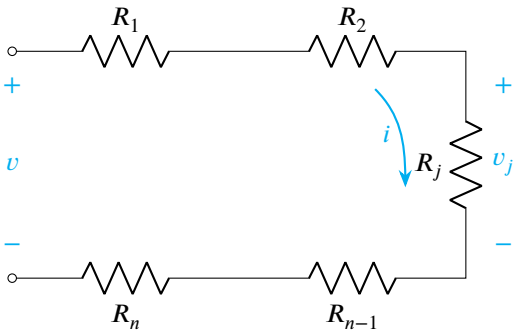
2.7.2 Parllel

$$R_{eq} = \frac{1}{\sum^{k} n = 1 \frac{1}{R_n}}$$



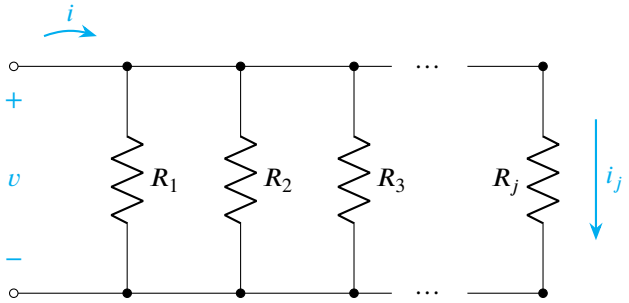
2.8 Voltage Division:

$$V_j = iR_j = \frac{R_j}{R_{eq}} V$$



2.9 Current Division:

$$i_j = \frac{V}{R_j} = \frac{R_{eq}}{R_j} i$$



2.10 Source Transformations:

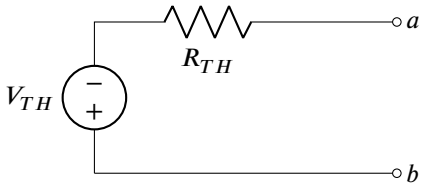
Current and voltage sources can be interchanged. This is because equivalent sources produce the same electrical effects, regardless of load.

2.10.1 Thévenin Equivalent

A way to simplify a circuit that focuses on the behaviour at the terminals.

$$V_{TH} = V_{O/C} = V_{ab}$$

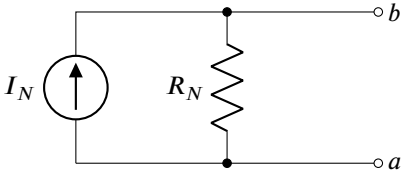
$$R_{TH} = \frac{V_{TH}}{I_{SC}}$$



2.10.2 Norton Equivalent

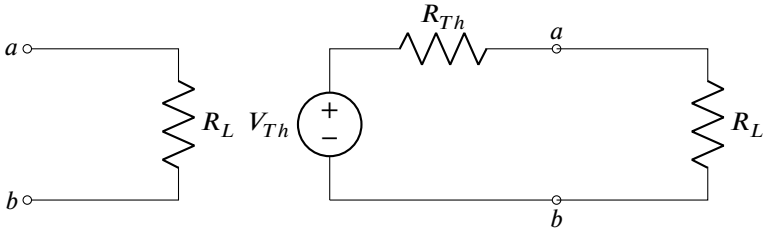
$$I_N = I_{S/C}$$

$$R_N = R_{TH}$$



2.11 Maximum Power:

To find a circuits maximum power, replace its resistive network within its Thévenin equivalent.



Power delivery to R_L:

$$p = I^2 R_L = (\frac{V_{TH}}{R_{TH} + R_L})^2 R_L$$

Max power is when $R_L = R_{TH}$:

$$p_{max} = \frac{V_{TH}^2 R_L}{(2R_L)^2} = \frac{V_{TH}^2}{4R_L}$$

2.12 Superposition Principle:

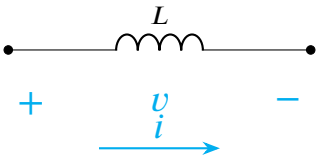
A linear system obeys this principle. Which is when two or more independent energy sources are connected in a circuit, the total response is the sum of each individual.

2.13 Inductors:

The behaviour of inductors are based around the properties of electro--magnetism. The SI unit for an inductor is Henry.

$$V = L\frac{di}{dt}$$

$$W = \frac{1}{2}LI^2$$



$$i(t) = \frac{1}{L}\int_{t_0}^{t} v\, d\tau + i(t_0)$$

$$p = Li\frac{di}{dt} = \frac{dw}{dt} = vi(t)$$

2.13.1 Series

$$L_{eq} = \sum_{n=1}^{k} L_n$$

2.13.2 Parallel

$$L_{eq} = \frac{1}{\sum_{n=1}^{k} \frac{1}{L_n}}$$

2.14 Capacitor:

Made-up of two conductors seperated by an insulator. It acts as an O/C to DC current and S/C to AC. An important note to remember that electrical charge is not directly transported through the capacitor.



$$V = IR$$

$$i = C\frac{dv}{dt}$$

$$v(t) = \frac{1}{C}\int_{t_0}^{t} i\, d\tau + v(t_0)$$

$$p = vi = Cv\frac{dv}{dt}$$

$$w = \frac{1}{2}Cv^2$$

2.14.1 Series

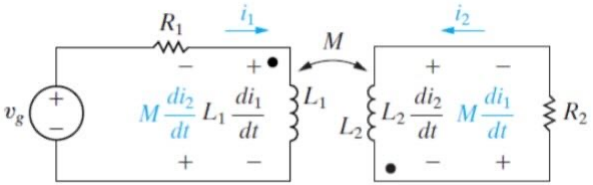
$$C_{eq} = \frac{1}{\sum_{n=1}^{k}\frac{1}{C_n}}$$

2.14.2 Parallel

$$C_{eq} = \sum_{n=1}^{k} C_n$$

2.15 Mutual Inductance:

The magnetic fields between the two circuits create a mutual and self--inductance voltages. Polarity of M depends on the way the coils are wound. L_1 and L_2 must be wound is opposing directions.


R_1
i_1
i_2
M
v_g
$M\frac{di_2}{dt}$
$L_1\frac{di_1}{dt}$
L_1
L_2
$L_2\frac{di_2}{dt}$
$M\frac{di_1}{dt}$
R_2